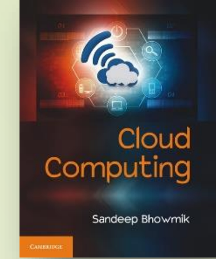
# Cloud Computing

Sandeep Bhowmik

## Chapter 8

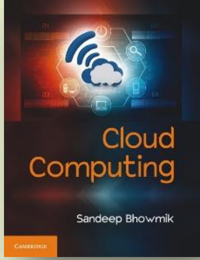
# Resource Pooling, Sharing and Provisioning

Cambridge University Press




# Resource Pooling

- In traditional computing people used to maintain discrete and independent set of resources, known as *silos*.
- On the other hand in cloud computing, consumers use well-connected pool of computing resources.
- The utility service model of cloud computing requires to maintain huge amount of all types of computing resources.
- Resource pooling in cloud needs setting up of strategies by provider for categorizing and managing resources.



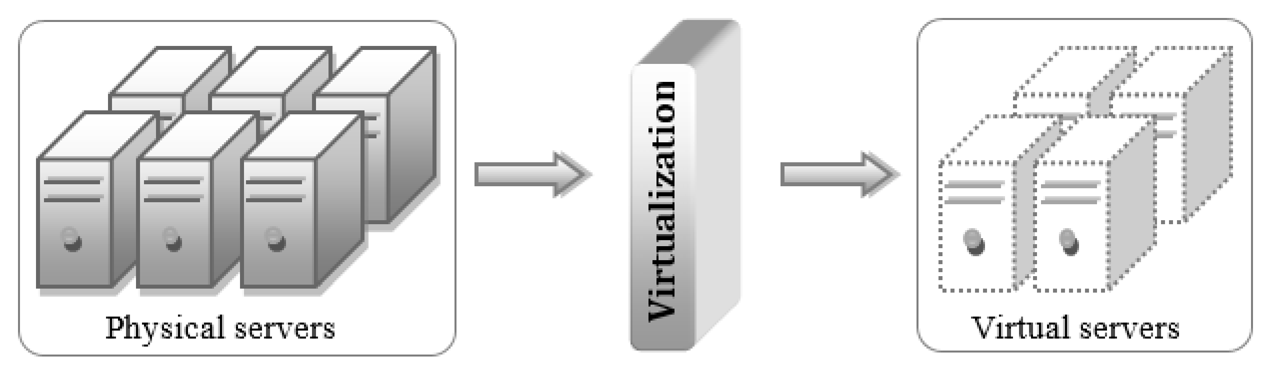

# Resource Pooling Architecture

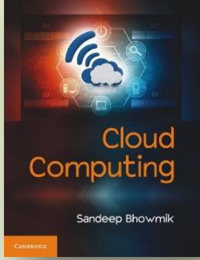
- A resource pooling architecture is designed to combine multiple pools of resources, where each pool groups together identical computing resources.
- The challenge is to build an automated system which will ensure that all the pools get together in synchronized manner.
- Computing resources can broadly be divided into three categories –
  - Computer/Server
  - Network
  - Storage.

# Computer or Server Pool

- Server pools are developed by building physical machine pools installed with operating systems and necessary system software.

- Virtual machines are built on these physical servers and combine into virtual machine pool.

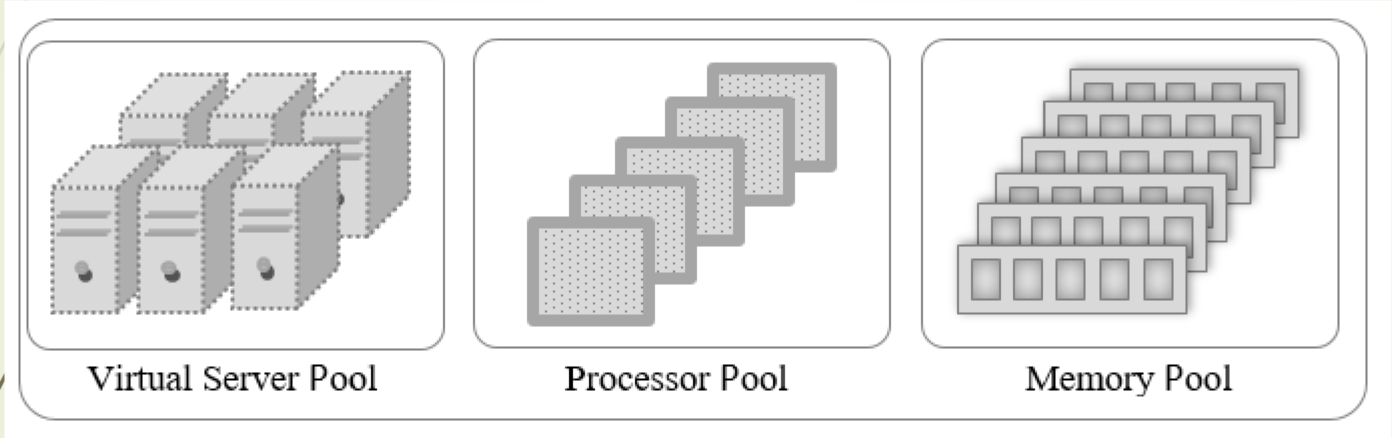





# Computer or Server Pool

- Dedicated pools of processor and memory are made.
- Processor and memory are allotted to virtual machines as and when required.
- Again they are returned to the pools of free components when load of virtual server decreases.

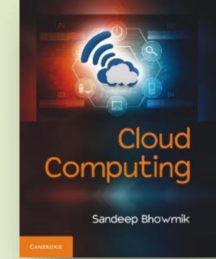
# Computer or Server Pool




Resource pool comprising of three sub-pools

# Storage Pool

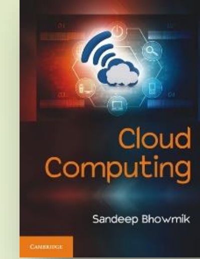
- Storage pools are made of block-based storage disks.
- They are configured with proper portioning, formatting and are made available to consumers in virtualized mode.
- Data stored into those virtualized storage devices are actually saved in these preconfigured physical disks.




# Network Pool

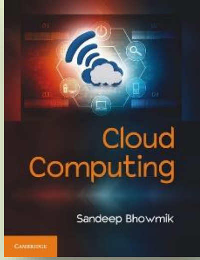
- Pool of physical network devices are also maintained in data centers.

- Pools of networking components are composed of different preconfigured network connectivity devices like switches, routers etc.



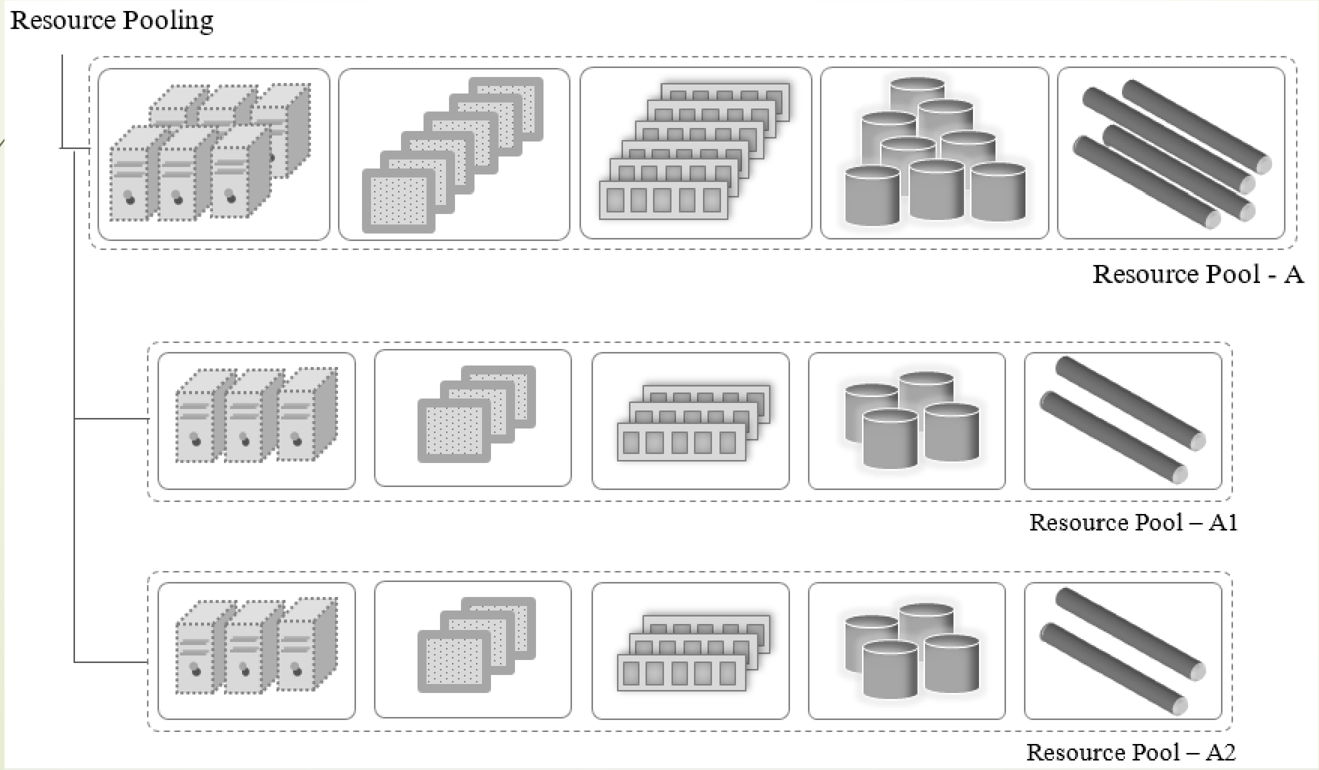

# The Hierarchical Organization

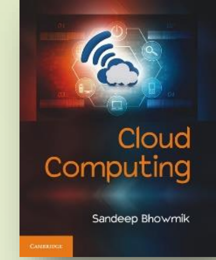
- A self-sufficient resource pool thus comprises of a number of smaller pools.
- Generally hierarchical structures are established to form parent, child relationships among pools.
- With such an organization, the pools eliminate the possibility of any single point of failure, as there are multiple similar resource components.



# The Hierarchical Organization

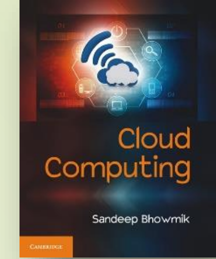
- Following is a sample hierarchical organization of pool, where pools A1 and A2 are comprised of similar type of sub-pools.





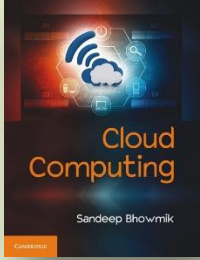

# Commoditization of the Data Center

- Commodity hardware is a device component that is -
  - widely available,
  - relatively inexpensive and
  - more or less interchangeable with other hardware of similar type.
- Cloud computing service promises to offer high-performance computing (HPC) facility.
- Technologists have been succeeded to produce this high computing performance by combining the powers of multiple commodity computing components.



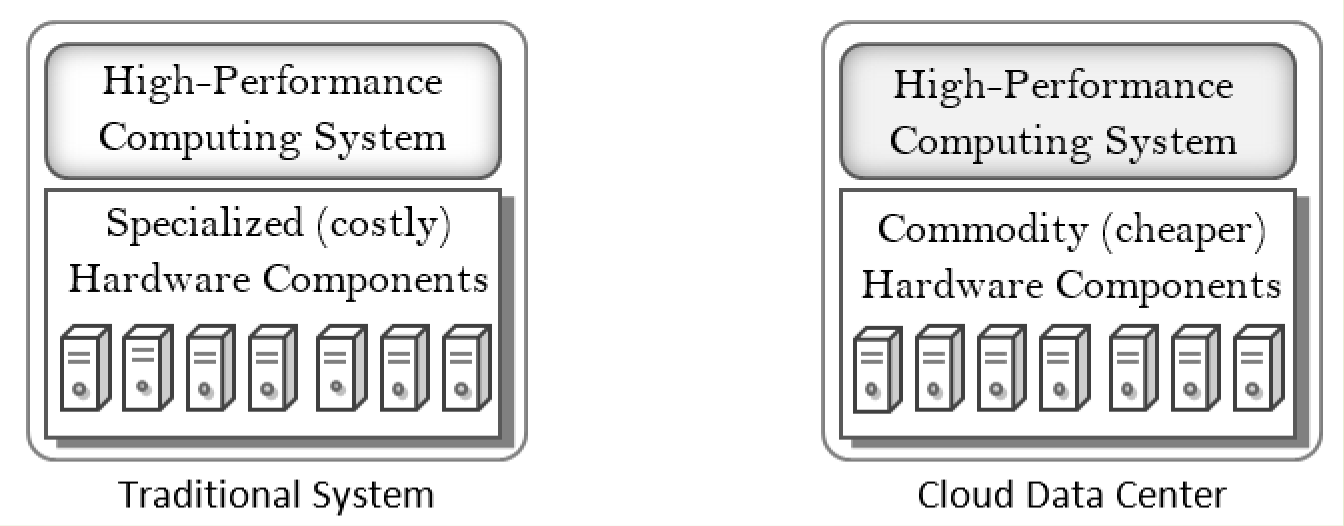

# Commoditization of the Data Center

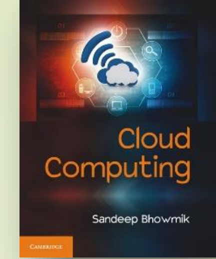
- Commoditization of resource pools at data centers has been an attracting feature of cloud computing.
- Commodity components are used to build the pools of (high performance) servers, processors, storage disks etc.
- As commodity components are cheaper than specialized components, commoditization of data centers is an important factor towards building widely acceptable cloud computing facilities.




# Commoditization of the Data Center

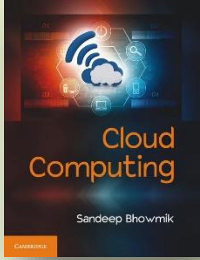
- Traditional system vs. commoditization of cloud data center.






# Commoditization of the Data Center

- Unlike commodity computing components, specialized components may not be easily available everywhere, all the time.
- The commoditization of data centers are also inspired by the diverse physical location of data centers on which clouds operate.
- With commoditization, it becomes easier for computing vendors to maintain the resource pools when replacement or new procurements are required.

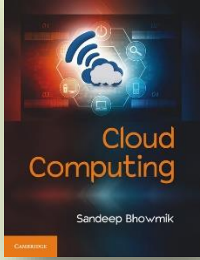



# Standardization, Automation, and Optimization

- Resource virtualization provide opportunities to –
  - Standardize,
  - Automate, and
  - Optimize

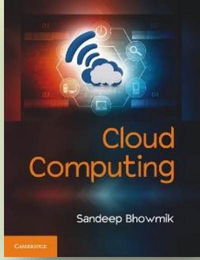
  deployments of cloud environment.

- These are the next set of essential activities for cloud data center development.



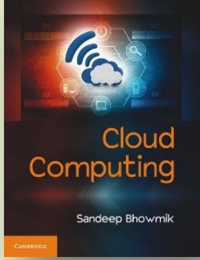

# Standardization

- Commodity resource components of a pool may come with various architectural standards.
- This creates need for simple and standardized logical representation of resources across the whole cloud environment.
- Standardization is done based on set of common criteria.




# Automation

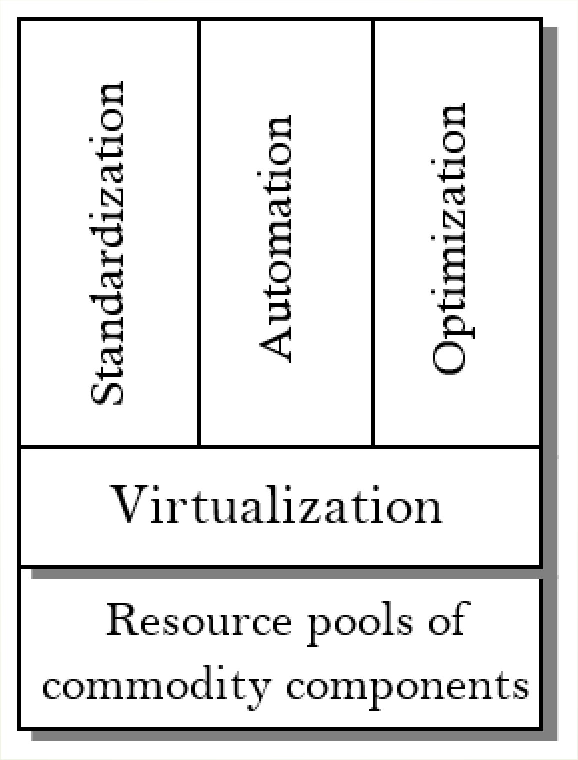
- Automation is implemented for –
  - resource deployment,
  - VM instantiation,
  - taking VMs off-line,
  - bringing them back online,
  - removing them rapidly and automatically
- based on previously set standard criteria.

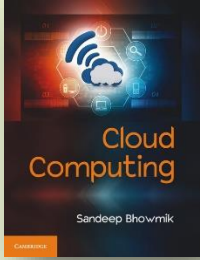



# Optimization

- This is done to optimize the resource usage to get –
    - optimal resource performance with limited set of resources.

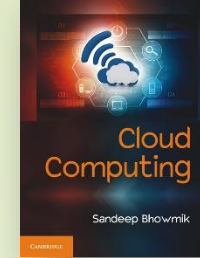
# Standardization, Automation, and Optimization





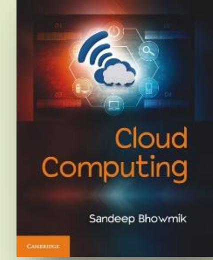

# Resource Sharing

- Resource sharing leads to higher resource utilization rate in cloud computing.
- Cloud computing allows sharing of pooled and virtualized resources among applications, users and servers.
- The average utilization of each resource component can be increased by sharing them among different applications, since all the applications do not generally attain their peak demands at same time.




# Resource Sharing

- Resource sharing implementation needs appropriate architectural support.
- The main challenge is to guarantee the Quality of Service (QoS).
- The sharing may affect the run-time behavior of other applications as multiple applications compete for the same set of resources.
- Thus, appropriate resource management strategies like optimization and decision based self-organization of system are required to maintain system performance.

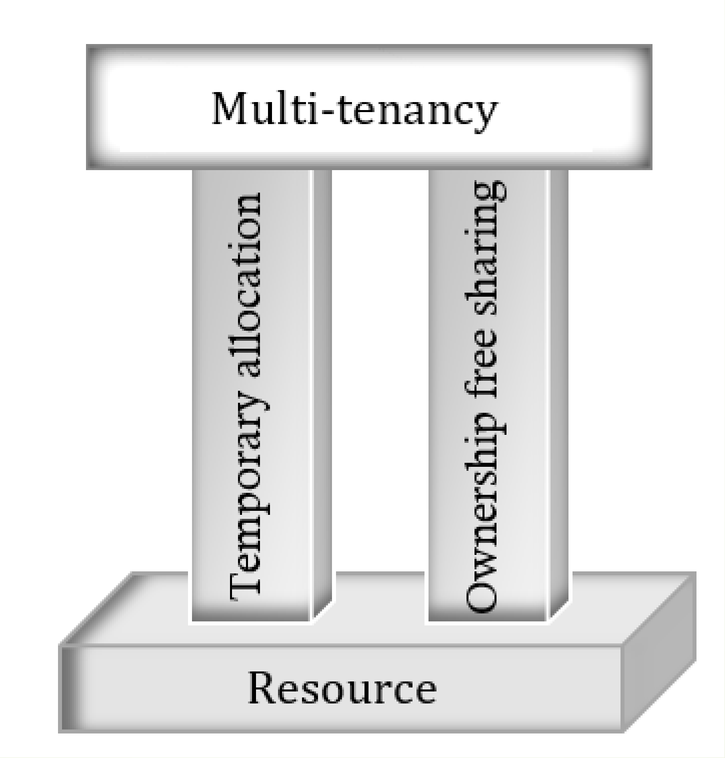


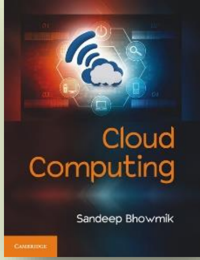

# Resource Sharing & Multi-tenancy

- Multi-tenancy is the –
  - characteristic of a system that enables a resource component to serve different consumers (tenants)
  - whereby each is isolated from the other

- Multi-tenancy in cloud computing has materialized based on the concepts of –
  - ownership-free resource sharing in virtualized mode
  - temporary allocation of resources from resource pool

# Resource Sharing & Multi-tenancy






# Resource Sharing & Multi-tenancy

- Cloud computing does not allow permanent acquisition of any resource component by any consumer.
- Multi-tenancy in cloud computing is largely enabled by the ideas of –
  - Resource sharing in virtualized mode
  - Dynamic resource allocation from resource pool

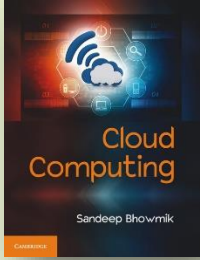
# Resource Sharing & Multi-tenancy

- **Types of Tenancy**

- Multi-tenancy in its actual sense allows co-tenants external to each other.

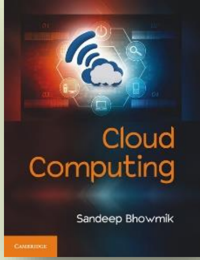
- Existence of external co-tenant is unique feature of public cloud.

- Private cloud only allows sub-co-tenants under one tenant.




# Resource Sharing & Multi-tenancy

- **Tenancy at Different Level of Cloud Services**
- Multi-tenancy works at all levels of cloud services.
- At IaaS level multi-tenancy provides shared computing infrastructure resources like servers, storages etc.
- At PaaS level multi-tenancy means sharing of operating system by multiple applications.
  - Applications from different vendors (tenants) can be run over same OS instance (to be offered as SaaS).




# Resource Sharing & Multi-tenancy

- **Tenancy at Different Level of Cloud Services**

- At SaaS level multi-tenancy refers to the ability of a single application instance and/or database instance to serve multiple consumers.

- Here, separation of data is logical only, although consumers gets the feeling of physical separation.

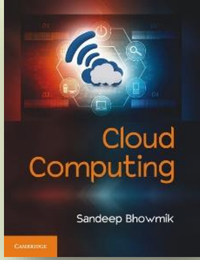
- Multi-tenancy eases application maintenance (like updates) and makes it economical for provider as well as consumers.

# Resource Provisioning

- Cloud provisioning is the allocation of cloud provider's resources to consumers.

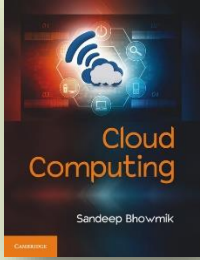
- Current age computing needs rapid infrastructure provisioning facility to meet the varying demands of consumers.

- With the emergence of virtualization technology and cloud IaaS model, it is now just a matter of minutes to achieve the same.



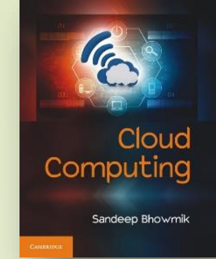

# Resource Provisioning

- Flexible resource provisioning is a key requirement in cloud computing.
- To achieve this flexibility, it is essential to manage the available resources intelligently.
- The orchestration of resources must be performed in a way so that resources can be provisioned to applications rapidly and dynamically in a planned manner.



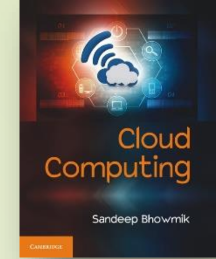

# Resource Provisioning

- **The Autonomic Way**
- *Autonomic resource provisioning* is an automated process in cloud which is designed applying artificial intelligence.
- The purpose of autonomic resource provisioning is to automate the allocation of resources.
- Through autonomic approach, computing resources can be rapidly provisioned and released with minimal management effort from a shared pool.



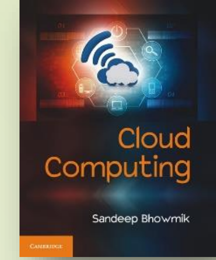

# Resource Provisioning

- **Role of SLA**
- Consumers typically enter into contract with cloud providers, to describe the expected requirements of computing resource capacity.
- This contract is known as *service level agreements* (SLA).
- Cloud providers estimate the resource requirements of consumers through the SLA contract that consumers make.



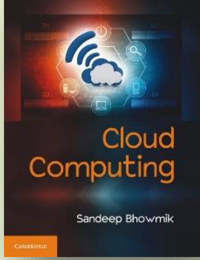

# Resource Provisioning

- **Resource Provisioning Approaches**
- There are two types of resource provisioning approaches - static and dynamic.
- In static approach, VMs are created with specific volume of resources and the capacity of the VM does not change in its lifetime.
- In dynamic approach, the resource capacity per VM can be adjusted dynamically to match work-load fluctuations.




# Resource Provisioning

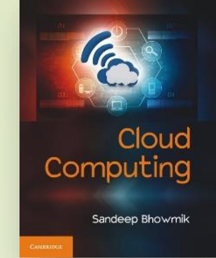
- **Resource Provisioning Approaches**
- In *static provisioning*, resources are allocated only once during the creation of VMs. This leads to inefficient resource utilization and restricts elasticity.
- In *dynamic provisioning*, resources are allocated as per application requirement during runtime.




# Resource Provisioning

- Comparison between static & dynamic resource provisioning.

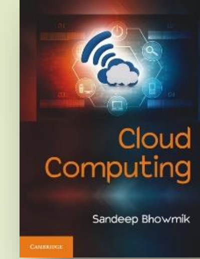
| Non-Virtualized Machine Environment | Virtualized Machine Environment |
|---|---|
| Resource allocation decision can be made once only for an application. | For an application, resource allocation decisions can be made number of times. |
| Resource allocation decision should happen before starting of the application. | Decision can be taken even after starting of the application. |
| This approach does not provide scope for elasticity to a system. | Provides scope for elasticity to the system. |
| Restrict scaling. | Enables scaling. |




# Resource Provisioning

- Comparison between static & dynamic resource provisioning.

| Non-Virtualized Machine Environment | Virtualized Machine Environment |
|---|---|
| Does not introduce any resource allocation decision overhead. | Incurs resource allocation decision overhead on system. |
| Resource once allotted can’t be returned. | Allotted resources can be returned. |
| Suitable when load pattern is predictable, and more or less unchanging. | Suitable for applications with varying workload. |
| Introduces resource under-provisioning and resource over-provisioning problems. | Resolves the resource under-provisioning and resource over-provisioning problems. |



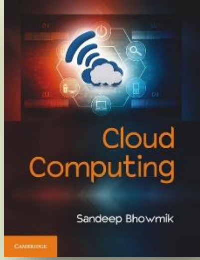

# Resource Provisioning

- **Hybrid Approach**

- Dynamic provisioning addresses the problems of static approach, but introduces runtime overhead.

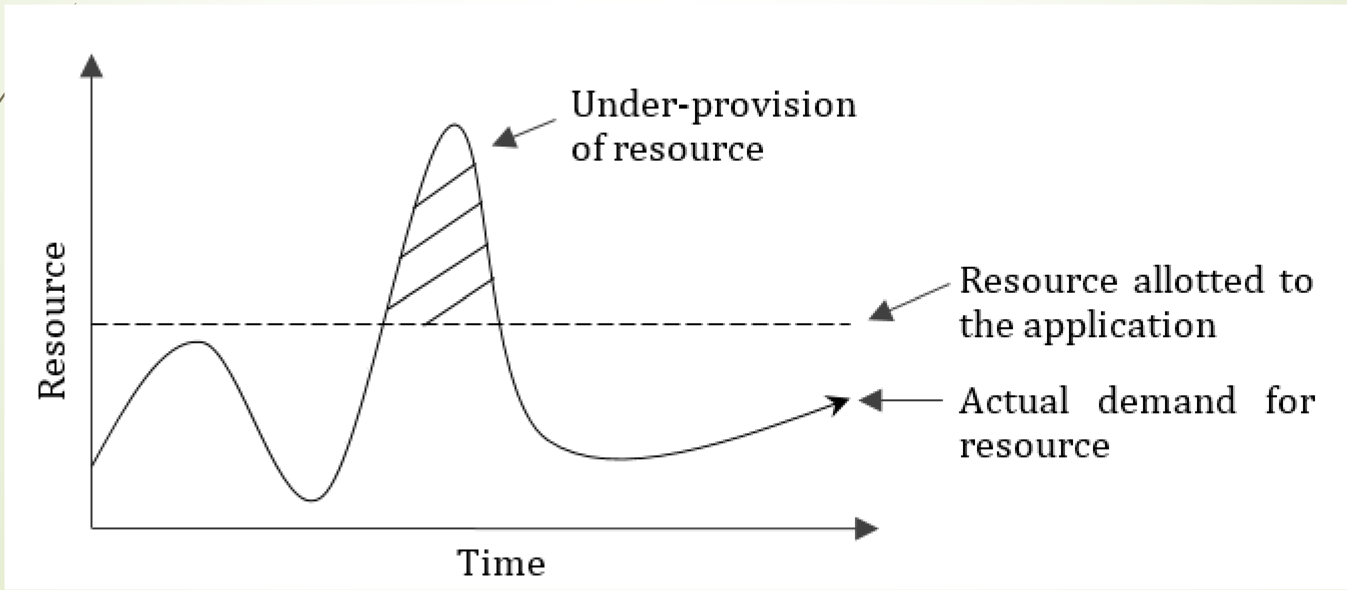
- To tackle this problem, a *hybrid provisioning* approach is suggested that combines both static and dynamic provisioning.

- It starts with static provisioning technique at the initial stage of VM creation, and then turns it into dynamic re-provisioning of resources.



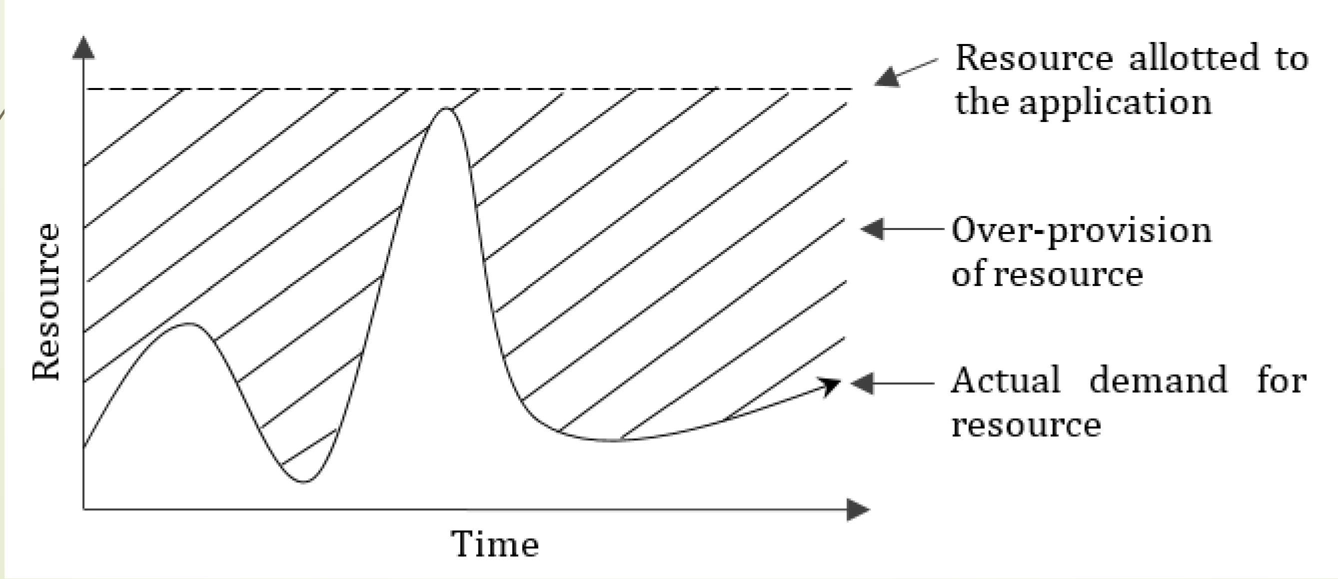

# Resource Under-provision & Over-provision Problems of Traditional Computing

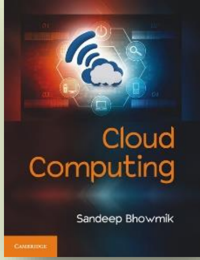
- Resource under-provisioning problem in traditional system



# Resource Under-provision & Over-provision Problems of Traditional Computing

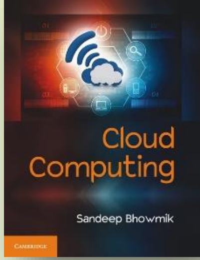
- Resource over-provisioning problem in traditional system






# Resource Under-provision & Over-provision Problems of Traditional Computing

- While over-provisioning wastes costly resources, under-provisioning degrades application performance and causes business loss.
- The static provisioning approach causes trouble for vendors also.
- Cloud computing addresses this issue by dynamic provisioning of resources using virtualization.




# Resource Provisioning Plans in Cloud

- Cloud providers generally offer two different resource provisioning plans or pricing models to fulfill consumers' requirements.
- The plans are known as *short-term on-demand plan* and *long-term reservation plan*.
- These two plans are made to serve different kind of business purposes.

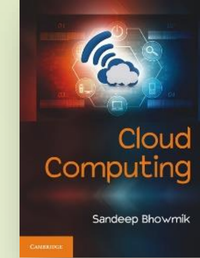
# Resource Provisioning Plans in Cloud

- **Short-Term On-Demand Plan**

- In this pricing model, resources are allotted on short-term basis as per demand.

- When demand rises, resources are provisioned accordingly to meet need. When demand decreases, the allotted resources are released returned to the free resource pool.

- Consumers are charged pay-per-use basis.



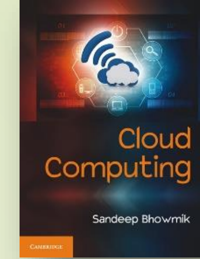

# Resource Provisioning Plans in Cloud

- **Long-Term Reservation Plan**

- In long-term reservation plan, a service contract is made between the consumer with the provider regarding requirement of resources.

- The provider then arranges in advance and keeps aside a certain volume of resources from the resource pool to support the consumer's needs.

- It is also known as advance provisioning.



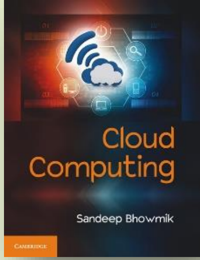

# Resource Provisioning Plans in Cloud

- **Long-Term Reservation Plan**

- Here, pricing is not on-demand basis. Rather it is charged as a one-time fee for a fixed period of time.

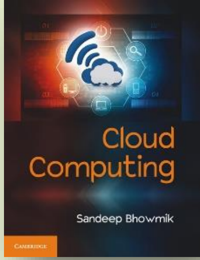
- The problem of the reservation plan is, since a fixed volume of resources are arranged as per the SLA, there are possibilities of under-provisioning or over-provisioning of resources.

- Application performance under this plan depends on consumer’s ability of estimating their own resource demand in advance.




# VM Sizing

- VM sizing refers to the estimation of the amount of resources that should be allocated to a virtual machine.
- Proper VM sizing leads to optimal system performance.
- VM sizing can be maintained in two different ways.

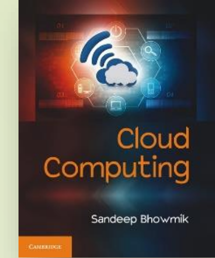



# VM Sizing

- Traditionally, VM sizing is done on a VM-by-VM basis, known as *individual-VM based provisioning*.

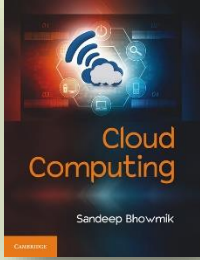
- In *joint-VM provisioning*, resources to VMs are provisioned in combined way, so that the unused resources of one VM can be allocated to other VM(s) when they are hosted over the same physical infrastructure.

- Joint-VM provisioning saves significant volume from overall resource capacity compared to individual-VM based provisioning technique.




# Dynamic Provisioning and Fault Tolerance

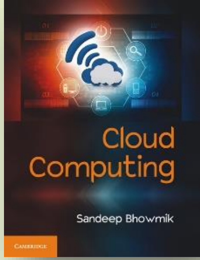
- Dynamic resource provisioning brings many advantages in cloud computing over traditional computing approach.
- It allows runtime replacement of computing resources and helps to build reliable system.
- It also turns the whole cloud system more tolerant to faults.




# Zero Downtime Architecture – Advantage of Dynamic Provisioning

- The dynamic provisioning mechanism replaces any crashing physical system with a new system instantly.

- This capability of cloud systems leads towards an important goal of system design – zero downtime architecture.




# Thank You